Final Year Project Proposal - Games Programming

1. Project Title

Narrative Framework Toolkit for Unreal Engine 5

2. Student Details

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Github project: https://github.com/users/laurenHBpencil/projects/3

3. Project Summary

This project will be making a reusable narrative development tool for Unreal Engine 5 aimed at making it easier to create story-driven games. The tool will hand developers with a set of tools/systems for managing story progression, player choices, dialogue and interactions without having to build these systems from scratch for every project which can be very time consuming.

This framework will use a story state system to store variables and player decisions. These values can then be used as conditions for dialogue, interactions and narrative events. The tool will be designed to work with both C++ and Blueprints, although it will mainly be in C++ and will include debugging features so developers can view and modify narrative states during testing. A small demonstration level will be created to showcase the framework.

4. Project Rationale

The reason I chose this project is because I have a big interest in story games, and I noticed that there seems to be a lack of simple tools for creating narrative systems. The tools I did find were too complex and included many features that were not necessary for what I wanted to create.

Over the summer, I was working on a narrative game and wanted to use an existing framework or tool to help manage the narrative systems. But the tool I found included a lot of functionality that I did not need, and I found myself wanting to recreate many of the systems myself in a

simpler way. So I spent more time learning their framework when it would've been easier for me to make myself.

This gave me the idea to make my own reusable narrative framework for Unreal Engine 5, focused on the core features needed for story-driven games without complexity, making development easier for designers or people with less experience.

5. Project Objectives

- Develop a reusable narrative framework for Unreal Engine 5.
- Create a system for storing and managing narrative variables and player choices.
- Develop a branching dialogue system that can react to stored story states.
- Create reusable interaction components that can be added to objects within a game.
- Allow developers to set conditions and actions for narrative events.
- A system for the saving and loading of narrative progress.
- Develop debugging tools for viewing and changing story variables during development.
- Create documentation explaining how another developer could use the framework.
- Produce a small demo project showing the main systems working together.

6. Technical Approach

Engine: Unreal Engine 5

Languages: C++ and Blueprint Visual Scripting

Project Type: Reusable Unreal Engine development tool/framework

Key Features:

- **Story State System** - stores values such as choices, events and progression using reusable IDs rather than game-specific code.
- **Branching Dialogue System** -dialogue options and responses can change depending on previous player choices.
- **Condition and Action System** -developers can define requirements for an event and actions that occur when those requirements are met.
- **Interaction Component** -reusable components that can be attached to actors to make them interactable.
- **Save/Load System** -stores narrative state between sessions.
- **Debugging Tools** -allows developers to inspect and modify narrative values while testing.
- **Blueprint Integration** -systems developed in C++ will be exposed to Blueprint so they can be easily used by designers.

The framework will be designed to be independent of any specific game. A small example project will be used only to demonstrate and test the tool.

High-Level Project Schedule

Development Phase	**Key Deliverable / Milestone**	**Target Date**
Framework Foundation	Core C++ structure and Blueprint integration	18 Dec 2026
Story State Management	Functional narrative state prototype	8 Jan 2027
Branching Dialogue	Dialogue and player choice system	22 Jan 2027
Conditions and Actions	Working conditional narrative logic	29 Jan 2027
Interaction System	Reusable interaction components	5 Feb 2027
Narrative Events	Functional narrative event triggers	12 Feb 2027
Save/Load System	Narrative progress saving and loading	19 Feb 2027
Debugging Tools	Narrative state inspection tools	26 Feb 2027
Integration and Refinement	Integrated and improved framework	12 Mar 2027
Demonstration Level	Playable framework demonstration	19 Mar 2027

Testing and Evaluation	Final testing and evaluation report	31 Mar 2027

7. Expected Outcome

The framework will let developers create branching conversations and store player choices. Events can be triggered based on previous choices, interactive objects can be created and the player's progress through the narrative can be saved.

A small level will be made to show off the features of the framework. Documentation will be given for setting up and using the toolkit, as well as a technical report detailing the design, implementation and testing of the framework.

8. Risk Management Assessment

Risk	Likelihood / Impact	Mitigation	Contingency
Scope creep	High / High	Focus on core narrative systems	Remove optional features
Development delays	Medium / High	Review progress regularly	Prioritise essential deliverables
C++ and Blueprint integration issues	Medium / High	Test throughout development	Simplify components
Save/load problems	Medium / High	Test narrative persistence frequently	Prioritise saving essential story variables

Data loss	Low / High	Use GitHub, Diversion and backups	Restore the latest working version
Insufficient testing	Medium / High	Test during each phase	Focus on functionality

Risks will be looked at through the whole development, especially at the end of each major stage. Any new risks will be found and changed based on impact. Where necessary, the project schedule or scope will be changed to prioritise the most important features and make sure the core framework is completed on time.

9. References

- **Story Framework by V1 Projects** - inspiration for the idea of a reusable framework for developing narrative games. https://www.fab.com/listings/108a7e42-a307-42fb-b1e9-e21a4c1905d7?lang=en
- **Unreal Engine 5** – development engine.
- **Epic Games Unreal Engine Documentation** - reference for C++, Blueprints, Actor Components, Data Assets and SaveGame systems.
- Narrative driven games such as Life is Strange as examples of the types of games that could make use of the framework.